

Timothy Guild

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Education

University of Arizona, Tucson, AZ - Mechanical Engineering, Dec. 2005 ➡ Major GPA : 3.10

Experience

BECHTEL - NUCLEAR WASTE TREATMENT PLANT, RICHLAND, WA ★ DURATION = 18 MONTHS TO PRESENT

- Chilled Water System Engineer - Pump & Line Sizing, Slug Flow Analysis, Process and Integration
- Nuclear Confinement Ventilation Design
 - Fluid Flow Modeling using FATE 2.0 - Normal and Off-Normal Conditions, Accident Analysis & Prevention
 - Full Facility Computer Ventilation Simulation

ESG ENGINEERING (CONSULTING FIRM), TEMPE, AZ ★ DURATION = 15 MONTHS

- Creative Conceptual Design - Napkin Sketch ➡ CAD Assembly ➡ Manufacturer Drawings ➡ Process Flow ➡ Prototype Build
- Component Optimization - Weight and Cost Reduction, Ease of Manufacturability, Software Derived
- Linear / Nonlinear Analysis - Assembly Crash Testing, Part Forming Residual Stress, Weld Stress, Drop Test, etc.
- Software -
 - Solidworks - Parts, Assemblies, Engineering Drawings
 - Altair Hypermesh - Finite Element Pre-processor
 - Altair Optistruct - Shape and Size Optimization, Linear Solver
 - LS-DYNA - Transient Dynamic Finite Element Program
- Products - Innovative Ambulance Seat ➡ Infant Cradle, Heavy Truck and Bus Seat Base Assembly, Trucking Coupler Redesign, Stirling Powered Solar Dish Design, Misc. Consumer Products, Etc.

G & B PRODUCTIONS, TUCSON, AZ ★ DURATION = 20 MONTHS

- Created Clothing Company Startup
 - The Business Model
 - The Product
- Compiled a Team to Develop, Market, Manufacture and Distribute a High End Clothing Line
- Software -
 - Adobe Photoshop - 3D Image Creation and Rendering
 - Adobe Illustrator - Image Creation and Experimentation

ORBITAL SCIENCES CORPORATION, CHANDLER, AZ ★ DURATION = 4 MONTHS / SUMMER 04 INTERNSHIP

- Thermal Analyst Intern
- Rocket Thermal Ablation Analysis Based on Projectile and Path
- Component-based Thermal Packaging
- Software -
 - Lanmin - NASA Derived Thermal Protection System Design for Spacecrafts
 - SINDA/G - Advanced Transient / Steady-state Thermal Modeler with Fluid Flow and Radiation.

UNIVERSITY OF ARIZONA HEAT TRANSFER LAB, TUCSON, AZ ★ DURATION = 22 MONTHS

- Researcher
- Liquid Cooling Electronics Packaging
 - Simplified Math Modeling Used in Conjunction with Experimental Data
- Experience with Various Lab Hardware Including Flukes, HP Data Acquisition / Control Units, Recirculation Systems, PID controllers, etc.
- Software -
 - Matlab - High Level Technical Computing
 - Labview - Acquire, Analyze, and Present Data

PARK PLACE CONDOMINIUMS - LEASING AGENT, TUCSON, AZ ★ DURATION = 24 MONTHS

- Managed Leasing Agreements for Approximately 100 Current Residents as well as Future Resident Marketing
- Required Excellent Management and Communication Skills